Secuura x VIT POC

Project Charter / Statement of Work

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1. Executive Summary

1.1 Purpose of Document

This Project Charter formally authorizes the initiation of the Secuura x VIT Proof of Concept (POC) project. It defines the project's objectives, scope, high-level solution, key milestones, and the roles and responsibilities of the involved parties. This document serves as the single source of truth for the project team and stakeholders, ensuring a shared understanding and commitment to the project's goals.

1.2 Project Name & Team Identification

Project Name: Secuura x VIT POC
Project Team ID: G05
Date: March 5, 2026

2. Business Case & Problem Statement

2.1 Background

Organizations across banking, HR, education, and government services frequently rely on documents (e.g., identity cards, academic certificates, regulatory records) to verify claims made by individuals. Current verification workflows are predominantly manual, requiring compliance analysts to visually inspect documents and cross-reference information. This process is not only slow and labor-intensive but also inconsistent and difficult to audit, creating operational bottlenecks and exposing organizations to fraud risk.

2.2 Business Problem

The core problem this project addresses is the **lack of a fast, reliable, and auditable method for verifying information contained within documents**. Traditional OCR tools only extract text; they do not validate the truthfulness of the extracted data. This forces verification teams to spend significant time manually reviewing documents, leading to:

- **High Operational Costs:** Valuable staff time is consumed by repetitive manual checks.
- **Inconsistent Results:** Manual inspection is subjective and prone to human error.
- **Poor Auditability:** Proving the accuracy of a verification decision after the fact is difficult.
- **Increased Fraud Risk:** Inconsistencies or fraudulent documents are harder to detect without automated cross-referencing.

2.3 Project Objectives

The primary objective of this POC is to build and validate a platform that automates the end-to-end document verification process, transforming it from a manual review into an explainable, trustworthy, and efficient workflow.

The specific goals are to:

- **Automate Field Extraction & Validation:** Develop an AI-driven pipeline that automatically extracts key data fields from uploaded PDFs and validates them against trusted external sources.
- **Provide Explainable Results:** Deliver verification results with visual indicators (color-coded highlights directly on the document) and detailed reason codes, fostering user trust and enabling quick decision-making.
- **Enhance Auditability:** Implement a comprehensive audit logging system to record all verification actions, providing a clear trail for compliance and regulatory reviews.
- **Reduce Manual Workload:** Demonstrate a significant reduction in the time and effort verification teams spend on document inspection.

3. Project Scope

3.1 In-Scope

- **Document Upload Interface:** A user interface for uploading PDF documents, including file format and size validation.
- **Backend API Service:** A central service to manage uploads, orchestrate the processing pipeline, and serve data to the frontend.
- **Document Processing Engine:** A service to extract raw text, layout information, and bounding box coordinates from uploaded PDFs.
- **AI Field Detection Engine:** Integration of an AI model (e.g., Google Gemini, Azure AI) to identify and extract key fields (e.g., Name, DOB, ID Number) from the text, along with confidence scores.
- **Canonical Mapping Layer:** A component to map detected, varied field names (e.g., "DOB", "Birth Date") to a standardized canonical schema (e.g., person.date_of_birth).
- **Validation Connector Layer:** Development of at least one connector to interface with a trusted external API (e.g., an identity registry) to validate extracted field data.
- **Verification Viewer:** A frontend PDF viewer that displays the original document with color-coded overlays highlighting the verification status of detected fields:
- **Green:** Verified against the external source.
- **Amber:** Verification uncertain, requires manual review.
- **Red:** Verification failed (mismatch or data not found).
- **Field Detail Panel:** A panel allowing users to click on highlighted fields to view the extracted value, validation status, reason code, and confidence score.
- **Audit Logging System:** A secure log to record all verification activities, including document ID, extracted fields, validation results, and timestamps.
- **Secure Document Storage:** A system for encrypted storage of uploaded documents with controlled access.

3.2 Out-of-Scope

- **Processing of Non-PDF Document Types:** The POC will only support PDF uploads. Support for images (JPEG, PNG) or other document formats is excluded.
- **Integration with Multiple External Data Sources:** The POC will integrate with **one** external validation API to prove the concept. A comprehensive library of connectors is not in scope.
- **Advanced Fraud Detection:** While the system will flag mismatches, advanced forensic analysis (e.g., signature verification, photo tampering detection) is not included.
- **User Management & Advanced Authentication:** The POC will assume a single user role. Complex user permissions, multi-factor authentication, and user administration features are out of scope.
- **Custom AI Model Training:** The POC will leverage pre-existing AI models or APIs. Training a custom, fine-tuned model is not part of this phase.
- **Mobile Application Development:** The user interface will be a web-based application only. Native mobile apps are out of scope.
- **Handwriting Recognition:** The system is designed for machine-printed text. Handwriting recognition is not within the scope of this POC.

4. High-Level Solution Overview

4.1 Core System Capabilities

The Secuura x VIT POC will deliver an intelligent document verification platform. It will allow users to upload a PDF, after which the system automatically processes it to extract, map, and validate key data fields against external, trusted sources. The results are presented in an intuitive interface, overlaying verification status directly onto the document for immediate understanding. The core promise is to move beyond simple text extraction to **field-level verification with visual proof and auditability.**

4.2 High level Architecture and Data Flow

The system follows a modular, service-oriented architecture as outlined in the Solution Architecture document. The high-level data flow is as follows:

1. **Upload:** A user uploads a PDF via the **Frontend Interface**. The file is sent to the **API Gateway/Backend Service** and stored securely in the **Document Storage**.
2. **Process:** The **Document Processing Engine** extracts text, layout, and bounding boxes from the PDF.
3. **Detect:** The **AI Field Detection Engine** analyzes the extracted text to identify key data fields (e.g., Name, DOB) and assigns confidence scores.
4. **Map:** The **Canonical Mapping Layer** standardizes the detected fields into the system's internal schema.

5. **Validate:** The **Validation Connector Layer** sends the standardized field data to a **Trusted External API**. The API response (verified, mismatch, or not found) is received.
6. **Audit:** All steps, from upload to validation result, are recorded in the **Audit Logging System**.
7. **Visualize:** The verification results are sent back to the **Frontend Verification Viewer**, which displays the PDF with color-coded highlights and allows users to inspect field-level details.

5. Project Milestones & Timeline

This POC is planned for a duration of 12 weeks.

5.1 Milestone 1: Project Initiation & Infrastructure Setup (Weeks 1-2)

- **Deliverables:**
  - Finalized project plan and detailed timeline.
  - Setup of development, staging, and production environments (e.g., cloud accounts, repositories).
  - Selection and procurement of access to external AI model(s) and validation API(s).
  - Technical design document review and approval.

5.2 Milestone 2: Core Development - Ingestion & Processing (Weeks 3-5)

- **Deliverables:**
  - Functional backend API service with document upload endpoint.
  - Implemented secure document storage.
  - Completed Document Processing Engine capable of extracting text and layout from a sample set of PDFs.
  - Basic audit logging of upload and processing steps.

5.3 Milestone 3: AI & Validation Integration (Weeks 6-8)

- **Deliverables:**
  - Integrated AI Field Detection Engine, successfully identifying key fields from processed documents.
  - Developed Canonical Mapping Layer to standardize detected fields.
  - Developed and integrated the first Validation Connector, successfully communicating with the chosen external API.

5.4 Milestone 4: Frontend Development & Visualization (Weeks 8-10)

- **Deliverables:**
  - Functional frontend application with Document Upload Interface and Document List Dashboard.
  - Integrated PDF Viewer with dynamic, color-coded overlays for verified (green), uncertain (amber), and failed (red) fields.

    - Implemented Field Detail Panel, displaying extracted value, validation status, confidence score, and reason code.
    - End-to-end data flow established from upload to result visualization.

5.5 Milestone 5: Testing, UAT & Deployment (Weeks 11-12)

- **Deliverables:**
    - Completion of system integration testing and bug fixes.
    - Successful User Acceptance Testing (UAT) with a small group of target users.
    - System deployed to the production environment.
    - Final audit log review capability demonstrated.
    - Final project documentation and handover.

6. Roles & Responsibilities

6.1 Project Governance

- **Project Sponsor:** (Executives from Secuura/VIT) Provides overall vision, funding, and high-level strategic direction. Removes roadblocks and champions the project.
- **Steering Committee:** Reviews progress against milestones, approves scope changes, and ensures the project aligns with business goals.

6.2 Core Team Roles

Each member has a primary focus, the success of the POC relies on high collaboration and shared responsibility for coding standards, documentation, and problem-solving.

- **Project Lead & Solutions Architect (Member 1):**
    - **Primary Responsibilities:** This member acts as the **Project Manager** and **Tech Lead** combined. They are responsible for bridging the business requirements (PRD) with the technical implementation.
    - **Specific Tasks:**
        - Facilitating daily stand-ups and ensuring the team stays on track with the 12-week timeline.
        - Maintaining the **Solution Architecture** integrity, ensuring all components (Frontend, Backend, AI) communicate effectively.
        - Managing the Git repository, code reviews, and CI/CD pipeline setup.
        - Acting as the primary point of contact for stakeholders (if applicable) and removing roadblocks for the team.
        - Mentoring team members on system design and integration points.

- **Backend & Integration Lead (Member 2):**
    - **Primary Responsibilities:** Owns the core logic of the server. This role covers the **API Gateway**, **Document Processing Engine**, and the **Validation Connector Layer**.
    - **Specific Tasks:**

        - Developing RESTful APIs for document upload and result retrieval.
        - Implementing the **Document Processing Engine** to extract text and layout from PDFs (using libraries like PyPDF2, pdfplumber, or similar).
        - Building the **Validation Connector** to interface with the chosen external API (e.g., REST calls to a mock or real identity service).
        - Setting up the **Secure Document Storage** logic (e.g., interfacing with cloud blob storage).

- **AI & Data Engineer (Member 3):**
    - **Primary Responsibilities:** Focuses on the intelligence of the system. This combines the **AI Field Detection Engine** and the **Canonical Mapping Layer**.
    - **Specific Tasks:**
        - Integrating the AI model (e.g., Google Gemini, Azure AI) to perform field detection from the extracted text.
        - Engineering prompts to ensure high accuracy in identifying fields like Name, DOB, and ID numbers.
        - Developing the logic for the **Canonical Mapping Layer** to standardize varied field names (e.g., mapping "DOB", "Date of Birth", "Birth Date" to a single schema).
        - Calculating and returning the **confidence scores** for detected fields.
        - Collaborating with the Backend Lead to pass data to the validation layer.

- **Frontend & Visualization Specialist (Member 4):**
    - **Primary Responsibilities:** Owns everything the user sees and interacts with. This role covers the **UI/UX** implementation as described in the PRD.
    - **Specific Tasks:**
        - Building the **Document Upload Interface** and **Document List Dashboard** using a modern framework (streamlit or React).
        - Integrating a **PDF Viewer library** (e.g., Mozilla's PDF.js) that supports custom overlays.
        - Implementing the dynamic **color-coded highlighting** (Green/Amber/Red) over specific text fields based on backend verification results.
        - Developing the **Field Detail Panel** that appears when a user clicks a highlighted area, showing extracted value, status, and reason codes.
        - Ensuring the frontend is responsive and provides clear error messages.

- **QA, DevOps & Audit Specialist (Member 5):**
    - **Primary Responsibilities:** Ensures the product is reliable, secure, and traceable. This role combines **QA Engineering**, **Security/Compliance**, and **Audit Logging**.
    - **Specific Tasks:**

- **Testing:** Developing and executing test plans to ensure the product meets the "Reliability" criteria in the PRD (graceful failure, meaningful errors).
- **Audit Logging:** Implementing the **Audit Logging System** to record all verification actions, storing them securely.
- **Security:** Advising on secure coding practices and ensuring document storage encryption is correctly implemented.
- **DevOps:** Assisting the Project Lead with deployment, environment setup, and monitoring.
- **Documentation:** Maintaining a record of test results and system configurations.